

1. Administrative & Study Identification

Full Study Title: A Phase 2 Study to Evaluate the Efficacy and Safety of Ianalumab (VAY736) in Patients with Primary Immune Thrombocytopenia (ITP) Previously Treated with at Least a Corticosteroid and a Thrombopoietin Receptor Agonist (TPO-RA)
Short Title/Acronym: VAYHIT3 **Protocol Number:** CVAY736Q12201 **NCT Number:** NCT05885555 **Sponsor Name:** Novartis Pharmaceuticals
Investigational Product: Ianalumab (VAY736) **Phase of Development:** Phase II **Medical Monitor:** Novartis Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate the therapeutic efficacy, safety, and tolerability of ianalumab in adult patients with primary ITP previously treated with at least one corticosteroid and one TPO-RA. **Secondary Objectives:** To assess the quality of response (e.g., stable response), time to treatment failure, and the long-term safety profile.

2.2 Background & Rationale To address the unmet medical need for therapies in immune thrombocytopenia (ITP) that lead to sustained, safe platelet counts after treatment is discontinued. Ianalumab is an investigational, fully human monoclonal antibody targeting the B-cell activating factor receptor (BAFF-R). It has a unique dual mechanism of action: direct B-cell depletion mediated by antibody-dependent cellular cytotoxicity (ADCC) and inhibition of B-cell differentiation and survival via BAFF-R blockade.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Single-arm **Blinding:** Open-label **Randomization:** N/A

3.2 Planned Enrollment & Duration Target \$N\$: 41 participants
Number of Sites: International study with 24 participating centers
Estimated Study Start: 31-May-2023 **Estimated Primary Completion:** 31-Dec-2026

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Male or female participants aged 18 years and older with a confirmed diagnosis of primary ITP. 2. Platelet count < 30 G/L and assessed by the investigator as

needing treatment at screening. 3. Prior treatment with at least a corticosteroid ($\pm$ IVIG) and a TPO-RA, experiencing loss of response, insufficient response, no response, or intolerance to their last ITP therapy.

4.2 Exclusion Criteria 1. Diagnosis of secondary thrombocytopenia or prior splenectomy. 2. Neutrophil count $< 1000/\text{mm}^3$. 3. Treatment with a B-cell depleting therapy (e.g., rituximab) or anti-BAFF (e.g., belimumab) within 12 weeks prior to the first administration of ianalumab, or use of rescue medications (e.g., platelet transfusion) within 14 days.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Ianalumab 9 mg/kg administered once every 4 weeks for a total of 4 doses. (An optional second course is available for participants who achieved confirmed response and later lost it). **Route of Administration:** Intravenous (IV) infusion. **Duration of Treatment:** 16 weeks (initial core treatment) followed by safety monitoring up to 24 months post-last dose.

5.2 Concomitant & Prohibited Therapy Permitted: Concomitant therapy with corticosteroids and/or TPO-RAs is allowed if clinically indicated, provided patients were on a stable dose for at least 14 days before the first ianalumab infusion. **Prohibited:** Rescue therapies (e.g., platelet/blood transfusions, plasmapheresis) within 14 days prior to the first dose. Other non-corticosteroid immunosuppressants must undergo appropriate washout periods.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures	:---
:---	:---	Screening	Day -14 to 0 Informed Consent, Medical History, Baseline Platelet Labs ($< 30 \text{ G/L}$) Baseline
Day 0	First Ianalumab Infusion, Safety Labs, Concomitant Med Review	Interim	Weeks 4, 8, 12 Ianalumab Infusions (Doses 2, 3, 4), Efficacy Assessment (Platelet Counts) End of Study
Week 25	Primary Efficacy Analysis (Week 25 Day 1), transitioning into 24-month Safety Follow-up		

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Confirmed response (ConfR), defined as achieving a platelet count of $\geq 50 \text{ G/L}$ at 2 or more consecutive assessments at least 7 days apart between Week 1 and Week 25, in the absence of rescue treatment for ≥ 4 weeks prior to assessment and without starting a new ITP treatment.

Measurement Method: Central/Local Laboratory Platelet Count Analysis.

7.2 Secondary & Exploratory Endpoints - Quality of response (stable response rate). - Time to treatment failure. - Incidence and severity of Treatment-Emergent Adverse Events (TEAEs).

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Standard collection during active treatment and extensively monitored throughout the 24-month safety follow-up period. **Serious Adverse Events (SAEs):** Routine 24-hour reporting timeline following site awareness. **Data Monitoring Committee (DMC):** Independent safety review mechanisms governing interim analysis (IA) and overall study progression.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) population for primary efficacy outcomes; Safety Set for all patients receiving at least one dose of the investigational product. **Sample Size Justification:** $N=41$ adequately powered for a Phase II single-arm study evaluating the efficacy signal and safety profile in a heavily pre-treated, refractory ITP population. **Handling of Missing Data:** Missing platelet assessments due to the initiation of rescue therapy or a new ITP treatment will be strictly classified as non-responses.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Scheduled onsite and remote monitoring across the 24 participating international centers. **Audit Trail:** Details on eCRF locking, source data verification, and data clarification forms via standard Novartis quality audit mechanisms.

11. Study Identifiers & Governance

Primary ID: CVAY736Q12201 **Registry Number:** NCT05885555 **Sponsor/Lead Organization:** Novartis Pharmaceuticals **Study Title:** VAYHIT3 **Official Regulatory Title:** A Phase 2 Study to Evaluate the Efficacy and Safety of Ianalumab (VAY736) in Patients with Primary Immune Thrombocytopenia (ITP) Previously Treated with at Least a Corticosteroid and a Thrombopoietin Receptor Agonist (TPO-RA)

12. Clinical Framework (ITP Specific)

Primary Condition: Primary Immune Thrombocytopenia (ITP)
Diagnosis Threshold: Platelet count < 30 G/L **Medical Methodology:** B-cell activating factor receptor (BAFF-R) blockade and B-cell depletion **Optimization Target:** Confirmed platelet response (≥ 50 G/L)

13. Study Procedures & Methodology

Management Pathway: Standard clinical pathway for relapsed/refractory ITP **Education Protocol (HCP):** Ianalumab administration protocols, concomitant tapering guidelines, and safety event tracking **Education Protocol (Patient):** Bleeding symptom recognition, onsite training, and safety reporting instructions **Quality Audit Mechanism:** Efficacy tracking (platelet logs) & patient adherence monitoring

14. Observation & Follow-Up Schedule

Total Duration: 24 months (follow-up after the last dose) **Onsite Visit Cadence:** Baseline, every 4 weeks during the infusion phase, and Week 25 primary assessment **Remote Monitoring Frequency:** Every 4 weeks (Q4W) post-primary assessment **Checklist Review Interval:** Every 12 weeks (Q12W) during the long-term safety phase

15. Sign-off & Approval

Role	Name	Signature	Date	:---	:---	:---	:---
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]		Lead		
Statistician	[Name]	_____	[DD-MMM-YYYY]				